

## NORTH EQUATORIAL 2 BUOY 41041, USA

WMO: 997397

Lat: 14.500N

Lon: 46.000W

Elev: 3

StdP: 101.29

Time Zone: -3.00 (W03)

Period: 05-15

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                      | (o)  | (p)   |
| 2             | 23.1       | 23.6 | 17.2                          | 12.3 | 24.7 | 17.8 | 12.8 | 24.8 | 11.9                  | 24.2 | 11.5 | 24.3 | 8.2                      | 70   | 0.693 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 10            | 1.4                    | 28.6            | 25.6 | 28.5 | 25.5 | 28.3 | 25.3 | 28.0                | 28.1 | 27.7 | 27.8 | 26.9 | 27.5 | 6.3                  | 80   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 28.0                            | 24.2 | 28.1 | 27.7 | 23.6 | 27.7 | 26.8 | 22.4 | 27.4 | 90.0          | 28.1 | 88.0 | 27.6 | 84.7 | 27.4 | 29.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 11.1          | 10.4 | 10.0 | 21.8                       | 29.5 | 0.5                | 1.3 | 21.4                                               | 30.4 | 21.1       | 31.2 | 20.9       | 31.9 | 20.5       | 32.9 |    |    |
| (5)               |      |      | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 25.9   | 25.1  | 24.5  | 24.7  | 25.0  | 25.6  | 26.0  | 26.2  | 26.8  | 27.3  | 27.3  | 26.8  | 25.9  |  |  |
|                                                                        | DBStd       | 1.14   | 0.66  | 0.62  | 0.71  | 0.73  | 0.75  | 0.64  | 0.58  | 0.61  | 0.57  | 0.60  | 0.64  | 0.63  |  |  |
|                                                                        | HDD10.0     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | HDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |  |  |
|                                                                        | CDD10.0     | 5819   | 467   | 406   | 455   | 450   | 485   | 478   | 503   | 521   | 518   | 537   | 504   | 494   |  |  |
|                                                                        | CDD18.3     | 2777   | 208   | 173   | 197   | 200   | 227   | 228   | 245   | 262   | 268   | 279   | 254   | 236   |  |  |
|                                                                        | CDH23.3     | 24317  | 1430  | 942   | 1078  | 1261  | 1811  | 1969  | 2283  | 2724  | 2949  | 3134  | 2658  | 2077  |  |  |
| (13)                                                                   | CDH26.7     | 2233   | 1     | 0     | 0     | 13    | 33    | 59    | 104   | 341   | 589   | 688   | 351   | 55    |  |  |
| (14)                                                                   | Wind        | WSAvg  | 7.2   | 8.1   | 8.2   | 7.3   | 7.1   | 7.5   | 7.9   | 7.5   | 6.4   | 5.7   | 6.5   | 7.3   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 543    | 21    | 6     | 8     | 6     | 5     | 13    | 34    | 77    | 103   | 126   | 94    | 50    |  |  |
|                                                                        | PrecMax     | 787    | 99    | 30    | 97    | 22    | 20    | 48    | 105   | 223   | 238   | 288   | 195   | 143   |  |  |
|                                                                        | PrecMin     | 304    | 3     | 1     | 1     | 0     | 0     | 3     | 2     | 14    | 25    | 15    | 30    | 1     |  |  |
|                                                                        | PrecStd     | 104    | 22    | 6     | 17    | 6     | 4     | 11    | 23    | 47    | 50    | 59    | 44    | 41    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 26.9  | 26.2  | 26.4  | 27.9  | 27.5  | 28.0  | 28.0  | 28.7  | 29.0  | 29.1  | 28.6  | 28.0  |  |  |
|                                                                        |             | MCWB   | 23.0  | 24.0  | 23.8  | 23.7  | 24.1  | 24.4  | 24.6  | 25.5  | 25.6  | 27.2  | 24.9  | 23.9  |  |  |
|                                                                        | 2%          | DB     | 26.5  | 25.8  | 26.3  | 27.0  | 27.4  | 27.6  | 27.9  | 28.4  | 28.6  | 28.6  | 28.3  | 27.6  |  |  |
|                                                                        |             | MCWB   | 22.6  | 23.2  | 23.6  | 23.5  | 24.0  | 24.2  | 24.5  | 25.4  | 25.6  | 25.7  | 24.7  | 23.7  |  |  |
|                                                                        | 5%          | DB     | 26.3  | 25.7  | 26.0  | 26.4  | 27.1  | 27.4  | 27.6  | 28.1  | 28.4  | 28.5  | 28.0  | 27.3  |  |  |
|                                                                        |             | MCWB   | 22.5  | 23.0  | 23.3  | 23.2  | 23.8  | 24.1  | 24.4  | 25.3  | 25.5  | 25.6  | 24.6  | 23.5  |  |  |
|                                                                        | 10%         | DB     | 26.1  | 25.6  | 25.7  | 26.0  | 26.9  | 27.1  | 27.3  | 27.9  | 28.2  | 28.3  | 27.9  | 26.9  |  |  |
|                                                                        |             | MCWB   | 22.4  | 22.7  | 22.9  | 22.9  | 23.7  | 24.0  | 24.3  | 25.2  | 25.4  | 25.5  | 24.5  | 23.3  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 25.3  | 25.8  | 25.8  | 25.7  | 25.0  | 25.7  | 26.0  | 28.0  | 28.4  | 28.6  | 25.8  | 25.0  |  |  |
|                                                                        |             | MCDB   | 25.3  | 25.8  | 25.9  | 25.8  | 27.0  | 27.0  | 26.8  | 28.1  | 28.4  | 28.6  | 27.4  | 26.8  |  |  |
|                                                                        | 2%          | WB     | 24.4  | 25.3  | 25.4  | 25.2  | 24.6  | 25.3  | 25.7  | 27.6  | 27.9  | 28.3  | 25.5  | 24.6  |  |  |
|                                                                        |             | MCDB   | 25.3  | 25.3  | 25.5  | 25.4  | 26.8  | 26.7  | 26.8  | 27.7  | 28.0  | 28.3  | 27.4  | 26.6  |  |  |
|                                                                        | 5%          | WB     | 23.9  | 24.8  | 24.9  | 24.7  | 24.4  | 25.0  | 25.4  | 26.8  | 27.6  | 27.8  | 25.2  | 24.3  |  |  |
|                                                                        |             | MCDB   | 25.3  | 25.2  | 25.4  | 25.5  | 26.6  | 26.7  | 26.8  | 27.4  | 27.7  | 27.9  | 27.3  | 26.5  |  |  |
|                                                                        | 10%         | WB     | 23.4  | 23.3  | 24.1  | 23.8  | 24.1  | 24.7  | 25.1  | 26.1  | 27.0  | 26.8  | 25.1  | 24.0  |  |  |
|                                                                        |             | MCDB   | 25.4  | 25.1  | 25.4  | 25.6  | 26.5  | 26.6  | 26.8  | 27.3  | 27.6  | 27.6  | 27.3  | 26.4  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 1.3   | 1.3   | 1.0   | 1.1   | 1.0   | 0.9   | 1.1   | 1.4   | 1.4   | 1.4   | 1.3   | 1.2   |  |  |
|                                                                        |             | MCDBR  | 1.3   | 1.3   | 1.0   | 1.5   | 1.0   | 1.1   | 1.3   | 1.5   | 1.6   | 1.6   | 1.5   | 1.5   |  |  |
|                                                                        |             | MCWBR  | 1.2   | 1.5   | 1.3   | 1.3   | 1.0   | 1.2   | 1.1   | 1.3   | 1.4   | 1.3   | 1.1   | 1.3   |  |  |
|                                                                        | 5% WB       | MCDBR  | 1.3   | 1.2   | 1.0   | 1.5   | 1.1   | 1.1   | 1.3   | 1.7   | 1.9   | 1.9   | 1.5   | 1.3   |  |  |
|                                                                        |             | MCWBR  | 1.2   | 1.9   | 1.5   | 1.3   | 1.0   | 1.1   | 1.1   | 1.7   | 1.8   | 1.8   | 1.3   | 1.3   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.394 | 0.418 | 0.406 | 0.439 | 0.486 | 0.572 | 0.553 | 0.492 | 0.473 | 0.414 | 0.399 | 0.404 |  |  |
|                                                                        | taud        |        | 2.479 | 2.373 | 2.441 | 2.356 | 2.196 | 1.962 | 2.023 | 2.243 | 2.309 | 2.519 | 2.544 | 2.487 |  |  |
|                                                                        | Ebn at Noon |        | 902   | 893   | 910   | 873   | 820   | 743   | 760   | 817   | 837   | 886   | 891   | 881   |  |  |
|                                                                        | Edh at Noon |        | 107   | 123   | 118   | 128   | 148   | 185   | 175   | 142   | 133   | 105   | 100   | 104   |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 5.17  | 5.92  | 6.54  | 6.80  | 6.65  | 6.49  | 6.54  | 6.54  | 6.19  | 5.44  | 5.08  | 4.92  |  |  |
|                                                                        | RadStd      |        | 0.26  | 0.24  | 0.21  | 0.18  | 0.28  | 0.23  | 0.13  | 0.21  | 0.27  | 0.25  | 0.28  | 0.28  |  |  |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (0 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page